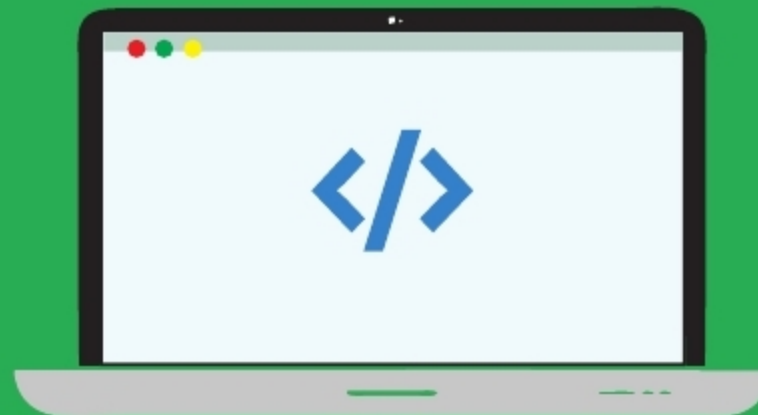


The Best Open Source IDEs *for* Windows Developers

The use of IDEs has become a part of the best practices adopted for developing software applications. IDEs can greatly enhance the coding experience and aid the coding process for software development by providing various facilities like code assist and debugging.



An IDE (Integrated Development Environment) is used to develop software applications. It can completely change the coding experience for a developer while creating a software application, yet we often do not realise the significance of its role. It facilitates coding by providing different features like intelligent code auto-completion, debugging, refactoring, source control, etc. All these save developers from doing mundane tasks and hence saves them time as well. IDEs optimise the software development work by automating different recurring tasks and simplifying the initial processes.

It was in the 1980s that developers started talking about the software development environment. Since then, software development has really advanced. The first tool to give a structure or framework to the software development process – for design, programming, performance or software maintenance – was designed back then. Subsequently, development tools were designed to meet the different needs that emerged in the world of software development. These tools, which were governed by a set of best practices to guide smooth programming, were grouped together as ‘computer aided software engineering’ tools. There have been continuous improvements in these tools since then. They are now oriented towards specific tasks to help developers in other programming environments as well. This is what we now refer to as the Integrated Development Environment (IDE).

An IDE is a set of tools integrated together to increase the productivity of software programmers. It generally includes a text editor that is used for programming, different functions that allow us to start the linker or compiler to translate the application source code into a machine readable binary set, and an online debugger that allows developers to run the current program, line by line. All software developers generally need these tools; hence, they are integrated together to optimise them. They have become very popular in business with particular IDEs designed by Google, Microsoft and Amazon to support their internal needs and adapt to their specific languages.

Now let us consider some open source IDEs that are popular among developers working on Windows.

Eclipse

This is a widely used free and flexible open source IDE that is useful for both beginners and professionals. Originally, Eclipse was a Java environment, but now it has got a wide range of capabilities because of the availability of a large number of extensions and plug-ins. Apart from various debugging tools and CVS/Git support, the standard edition of Eclipse also comprises Java and plugin development tooling. That’s not all — there are plenty of other packages available like tools for modelling, charting, testing, reporting and